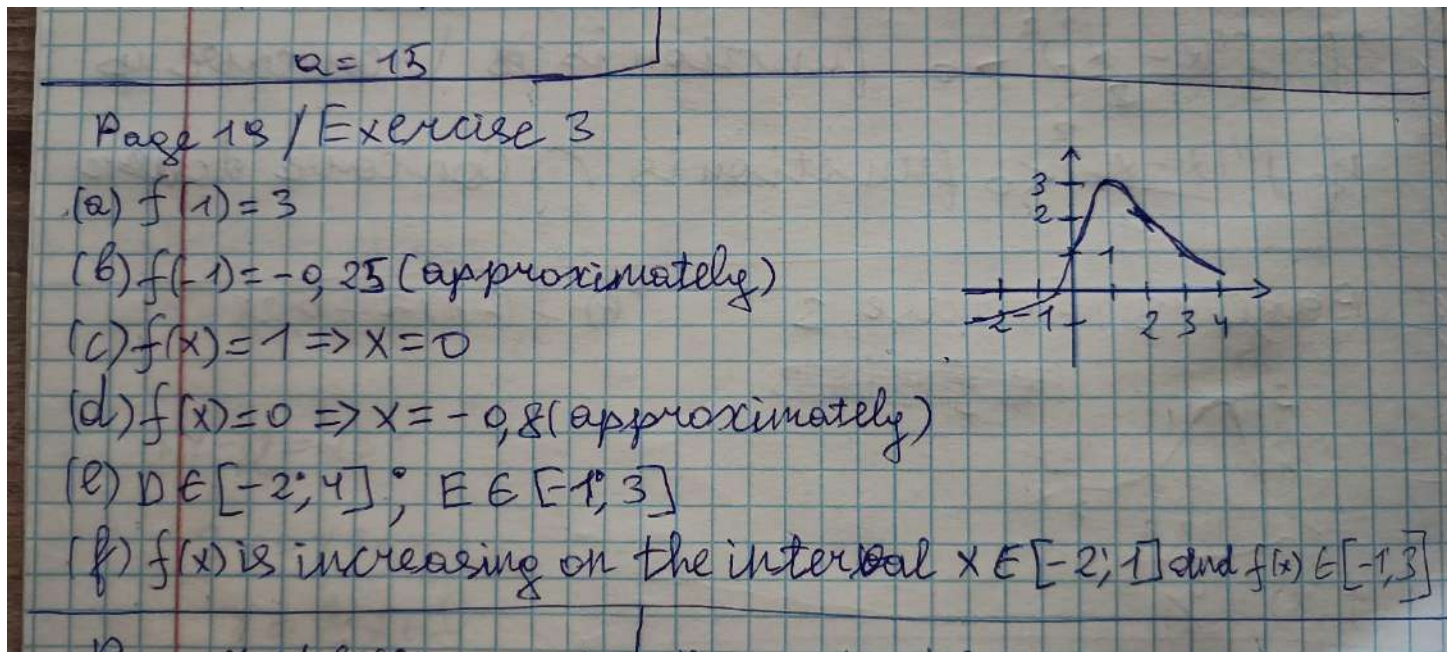


Javohirbek Xatamov

U2210251

Page 19/ ex 3



Page 71/ ex 63

Page 71 / exercise 63

$$\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$$

$$x^2 + x - 2 = 0$$

$$x = -2$$

$$(-2)^2 - 2 - 2 = 0$$

$$0 = 0$$

$$3x^2 + ax + a + 3 = 0$$

$$3(-2)^2 - 2a + a + 3 = 0$$

$$12 + 3 = a$$

$$a = 15$$

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2 \\ ax^2 - bx + 3 & 2 \leq x < 3 \\ 2x - a + b & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{x-2} =$$

$$\lim_{x \rightarrow 2^+} ax^2 - bx + 3 = 4a - 2b + 3 = 4$$

$$\lim_{x \rightarrow 3^-} ax^2 - bx + 3 = 9a - 3b + 3$$

$$\lim_{x \rightarrow 3^+} 2x - a + b = 6 - a + b$$

$$9a - 3b + 3 = 6 - a + b$$

$$10a - 4b - 3 = 0$$

$$4a - 2b + 3 = 4$$

$$4a - 2b - 1 = 0$$

$$\begin{cases} 8a - 4b - 2 = 0 \\ 10a - 4b - 3 = 0 \end{cases}$$

$$-2a + 1 = 0 \Rightarrow a = 0,5$$

$$\frac{4 \cdot 0,5}{2} - 2b = 1$$

$$2b = 1 \Rightarrow b = 0,5$$

$$a = b = 0,5$$

Page 111 / exercise 7

$$y = \sqrt{x}; \text{ ~~scribbles~~ }$$

$$f(x) = \sqrt{x}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$y - f(x_0) = f'(x_0)(x - x_0)$$

$$(x_0, y_0) = (1, 1)$$

$$f'(1) = \frac{1}{2}$$

$$y - 1 = \frac{1}{2}(x - 1)$$

$$y = \frac{1}{2}x - \frac{1}{2} + 1 \Rightarrow y = \frac{1}{2}x + \frac{1}{2}$$

Page 161 / exercise 21

$$f(1) = 2; \quad f'(1) = ?$$

$$f(x) + x^2 \cdot [f(x)]^3 = 10$$

$$f'(x) + 2x \cdot [f(x)]^3 + x^2 \cdot 3 [f(x)]^2 \cdot f'(x) = 0$$

$$f'(1) + 2 \cdot 1 \cdot [f(1)]^3 + 1 \cdot 3 [f(1)]^2 \cdot f'(1) = 0$$

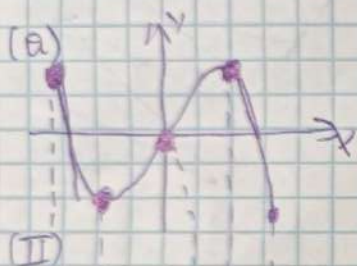
$$f'(1) + 16 + 12 \cdot f'(1) = 0$$

$$f'(1) = -\frac{16}{13}$$

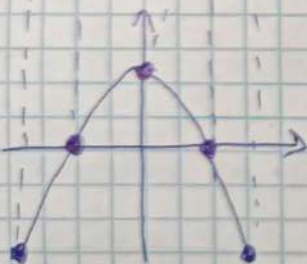
$$f(x+2) = 4$$

Page 122 / Exer 13

(a) - II



function (a) goes up when function II is above 0, and goes down when II is below 0, also ~~both of them~~ when II is equal to 0, and vice versa



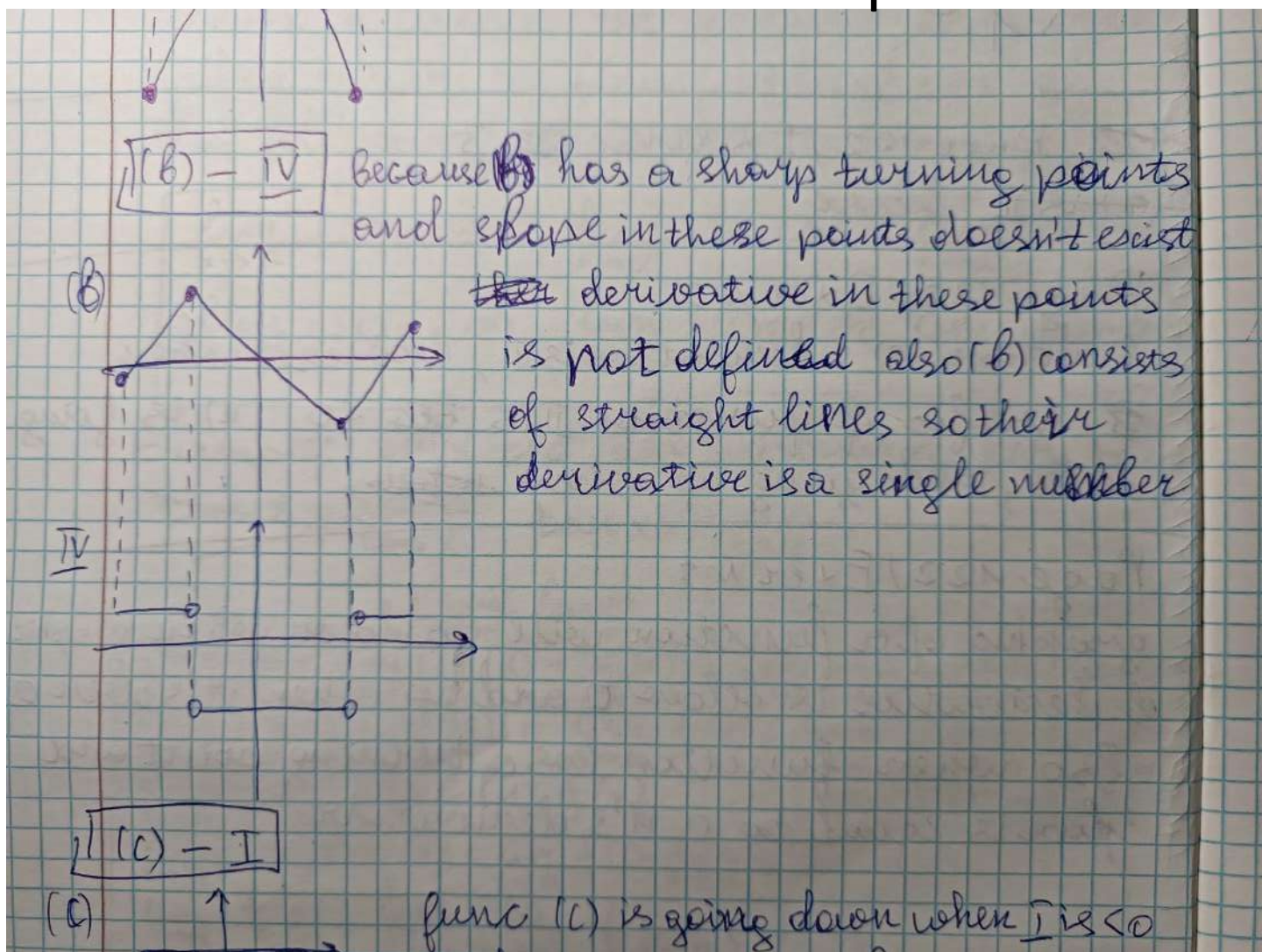
(b) - IV

(b)



because (b) has a sharp turning points and slope in these points doesn't exist ~~the~~ derivative in these points

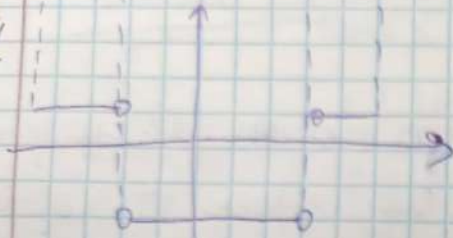
Second part



third part

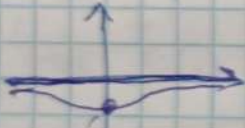
derivative is a single number

IV



$$f'(c) - I$$

(C)



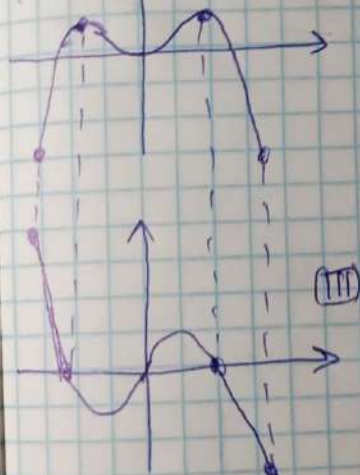
(I)



func $f(c)$ is going down when I is < 0
and up when $I > 0$, also when $I = 0$
 $f(c)$ has a turning point when
slope is $= 0$.

Fourth part

$$(d) - III$$



when (d) has a turning point (III) is equal to 0, also when $III > 0$, (d) goes up and down when $III < 0$.

Newton's method